

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · CHEMISTRY

CHEMISTRY 6027/2
PAPER 2 November 2023
2 hours 15 minutes

Candidate Name: _____

Centre Number: _____ **Candidate Number:** _____

TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES

Data booklet provided. Full structural formulas required for organic questions.

Answer all compulsory questions. Show clear physical chemistry derivations.

SECTION A

1. For the Haber Process equilibrium: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ [$\Delta H = -92 \text{ kJ/mol}$] (a) Write the expression for K_p including units when pressure is measured in atmospheres. (b) Explain the effect of increasing temperature on K_p and the equilibrium position. (c) Explain why industrial plants use 450°C and 200 atm rather than higher or lower values. *[Equilibria & K_c / K_p]* [8]
- (a) Write the expression and units for K_p . [2]
- (b) Explain the effect of temperature on K_p . [3]
- (c) Explain the economic compromise of 450°C and 200 atm. [3]